

AI-BASED NATIONAL SKILL MATCHING & MSME WORKFORCE PLATFORM

SELF EMPLOYMENT & TALENT UTILISATION

Pramaan.AI

The Standard of Proof, We just built the machine for it.

Turning invisible, undocumented skill into a verified, portable credential — so every worker can prove it, and every MSME can trust it.



Skill Credit Score



Reverse Skilling



Portable Credentials

THE PROBLEM

India's workforce skill is real — but invisible on paper

Millions of skilled tradespeople have no way to prove what they can actually do, and the small businesses hiring them have no reliable way to check.



No proof of skill

A welder or electrician's ability lives only in word-of-mouth — there is no verifiable, comparable record of what they can do.



MSMEs hire blind

Small manufacturers can't verify claims, so hiring is a gamble — costly trial-and-error on the shop floor.



Credentials aren't portable

Whatever trust a worker earns at one shop resets to zero the moment they look for the next job.

Widely-cited estimates put over 90% of India's workforce in the informal sector, spread across 60M+ MSMEs — a structural, national-scale gap.

One AI engine. Three problems solved at once.

Pramaan.AI is a full-stack platform where an AI examiner grades real skill, an AI trainer writes onboarding for new hires, and an AI matchmaker connects the two — end to end.



Skill Credit Score

A worker describes how they perform a task; the AI examiner grades it 0–900 like a real trade viva — vague answers score low, specific and safety-aware answers score high.



Reverse Skilling

An MSME owner describes their exact machine or SOP in plain language; the AI instantly writes a custom micro-training capsule — steps plus a quiz — for the next hire.

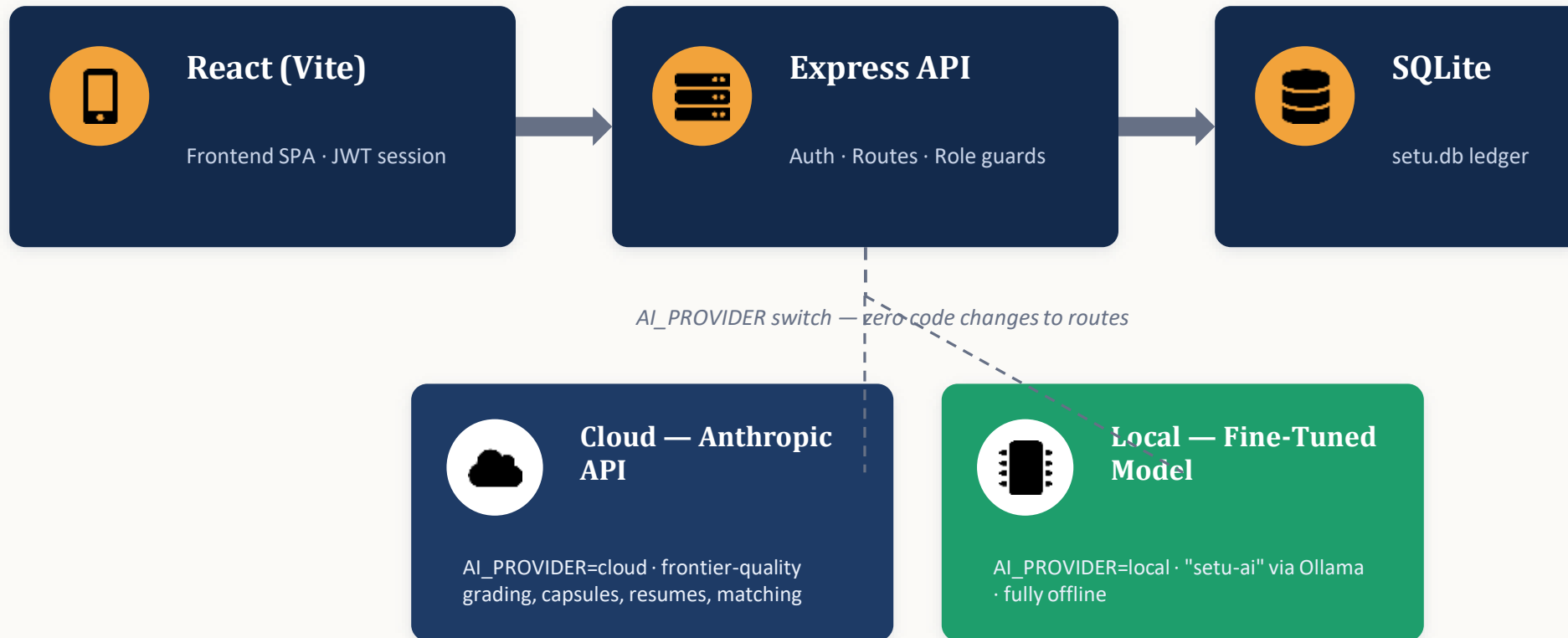


Verified Matching

The AI ranks the whole worker network against a job's real requirements — score, trade, location, language — with a plain-English reason for every match.

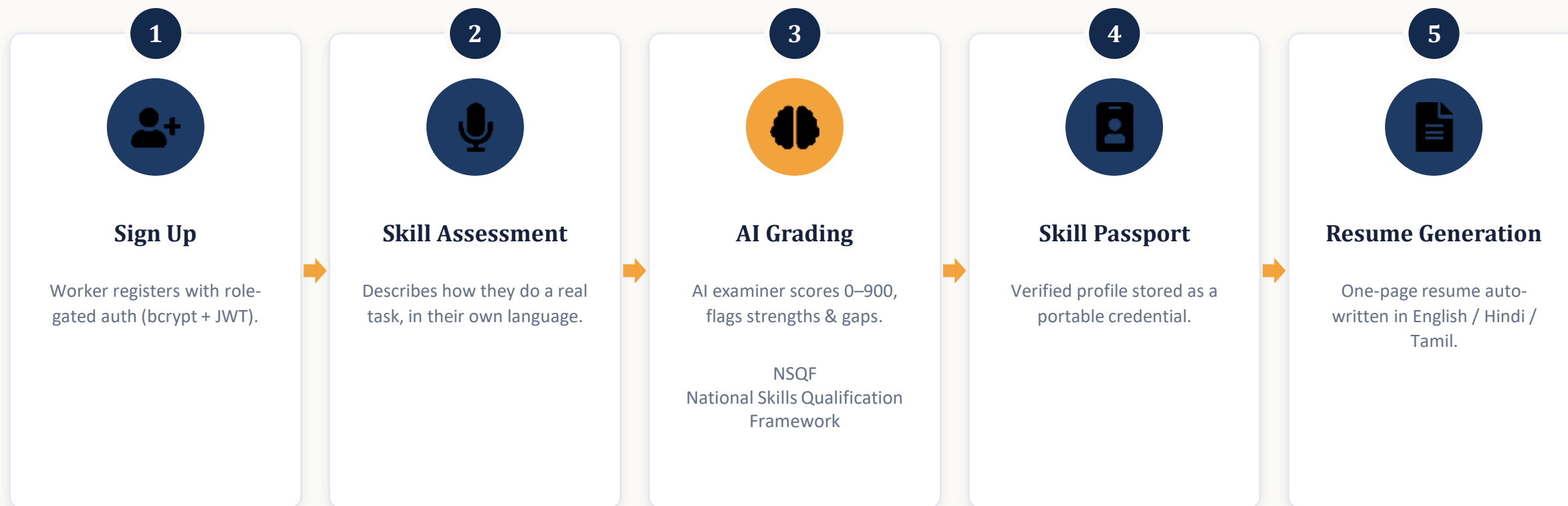
A real full stack — not a slide-ware prototype

Every AI call is brokered by the backend, so no credentials (or, in local mode, no data at all) ever reach the browser.



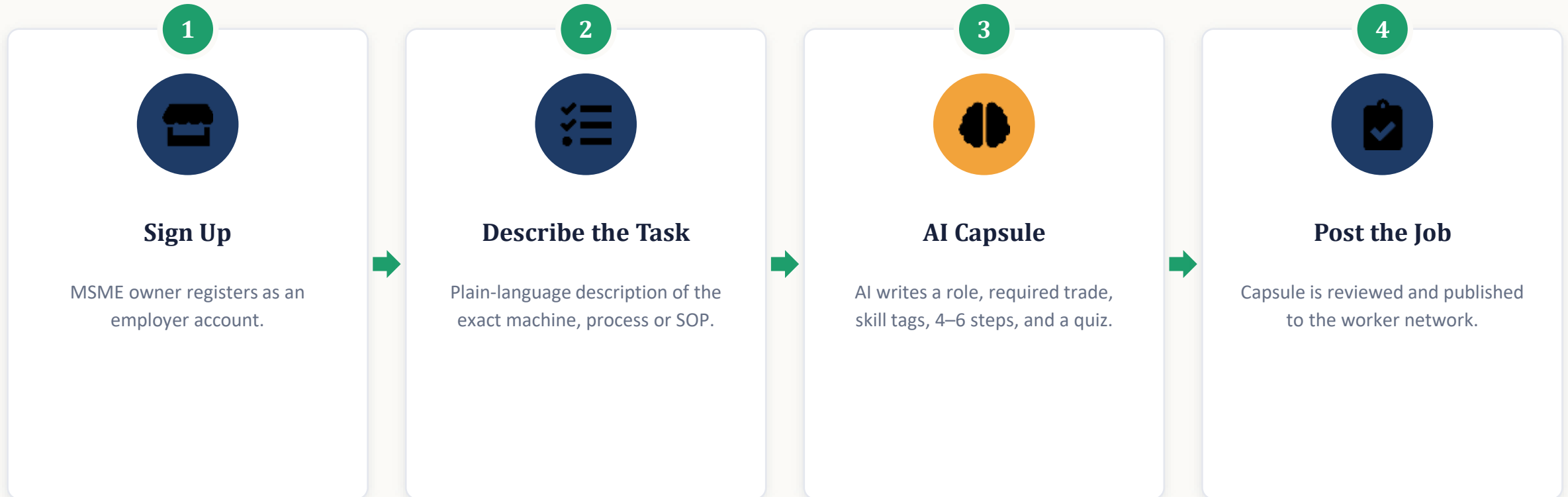
callClaudeJSON(system, prompt, maxTokens) — identical interface on both paths

The Worker Journey — from spoken skill to verified passport



AI grading is strict by design: vague, generic answers score under 400 — specific, safety-aware, technically correct answers score 700–900.

The MSME Journey — from a shop-floor task to a ready hire



Reverse skilling: every generated step traces back to the owner's exact words — never generic, boilerplate trade advice.

The AI Matching Engine — ranking the whole network, explained

Signals the AI weighs



Skill Credit Score, 0–900



Trade match to job requirement



Location proximity



Spoken-language fit



Sample ranked output — Job: Welder, Coimbatore

94

Muthu Raman

Score 810 · exact trade match · same city

81

Ravi Kumar

Score 760 · related trade · nearby district

68

Anil Mestri

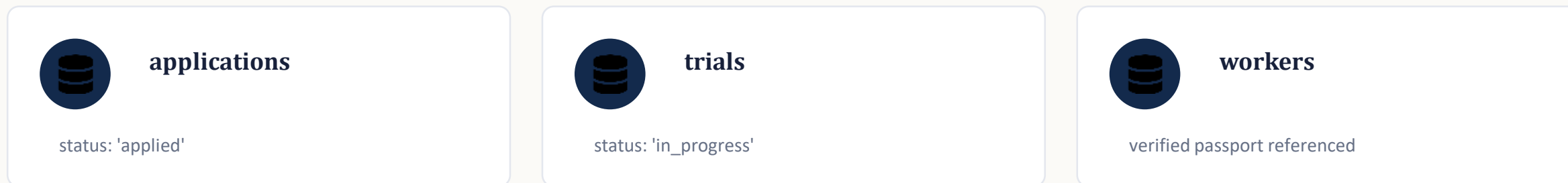
Score 640 · exact trade · language fit only

Model returns strict JSON: top 3 candidates, a 0–100 match score, and a one-line rationale citing score, trade and location/language fit.

From Application to Hire — a low-risk funnel for both sides



Every step is a real database record — not a mock



From cloud AI to Pramaan's own fine-tuned model

The same four tasks — grading, capsules, resumes, matching — can run entirely offline on a model fine-tuned on real usage.

1



Seed Data

13 hand-written examples covering all 4 tasks.

2



Live Logging

Real cloud calls logged as free training data.

3



Merge Dataset

Seed + logs merged into train.jsonl.

4



QLoRA Fine-Tune

Qwen2.5-3B tuned on a single consumer GPU.

5



Quantize & Export

GGUF export, registered into Ollama.

6



Serve Locally

"setu-ai" model — zero cloud dependency.

Both providers implement the identical `callClaudeJSON()` interface — switching is one `.env` line, zero code changes.

Simple, provable, and honest about its limits



React + Vite

Single-page frontend, i18n in 3 languages



Express

REST API, role-gated middleware



SQLite

better-sqlite3 — one file, zero services



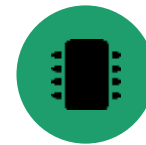
bcrypt + JWT

Hashed passwords, short-lived sessions



Anthropic API

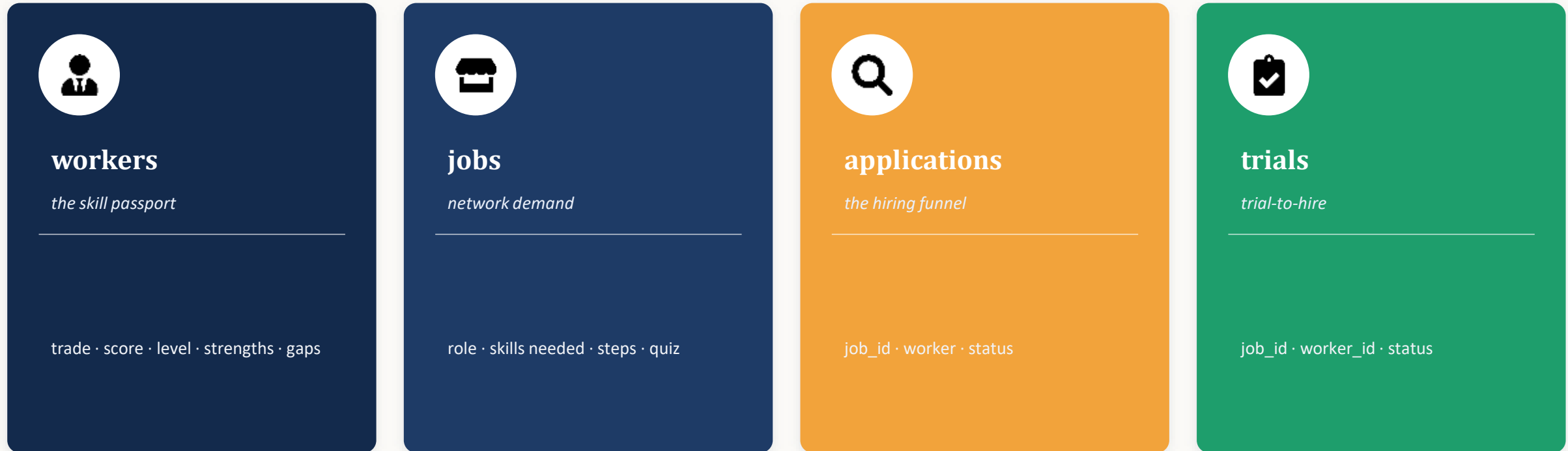
Cloud-grade grading & generation



Qwen2.5 + Ollama

Own fine-tuned model, fully offline

The database is the pitch, written as a schema



users → authenticates both roles (worker / msme) with bcrypt-hashed passwords and JWT sessions, enforced server-side.

Built to be trusted, not just demoed



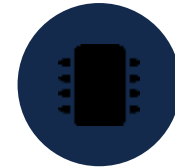
Real authentication

bcrypt-hashed passwords and configurable, short-lived JWT sessions — not hidden client-side checks.



Server-enforced roles

Worker vs MSME permissions are checked in Express middleware, never trusted from the UI alone.



Privacy by design

In local mode, worker data never leaves the server — a genuine offline option, not just a cost saving.

Honest about scope: hackathon prototype today — rate limiting, HTTPS termination and secret rotation are the next steps before production traffic.

Built for national scale, not a single city



90%+

of India's workforce is informal



60M+

MSMEs are the real hiring engine



3

languages served today — more
are one prompt away



0–900

point Skill Credit Score, common
across every trade

Every skill graded is a small, permanent gain for the worker — and every match made is one less blind hire for an MSME.

What ships next



Beckn-style network

Open, interoperable discovery so any registry can plug into the same match-and-hire protocol.



More languages

Every resume and capsule prompt is language-agnostic — adding a state language is a config change.



Offline-first mobile

Voice-based assessment on a basic smartphone, for workers with patchy connectivity.



Employer verification

MSME identity checks so a Skill Passport carries even more institutional trust.



Let's make skill visible.

Pramaan.AI — Proof of Skill for India's Workforce

Thank you — Happy to walk through a live demo.